**GE VERNOVA — GRID SOLUTIONS (PORTFOLIO SAMPLE)**

**TROUBLESHOOTING MANUAL**

# Circuit Breakers

TS-CB-2210 | Revision 01 | Effective Date: 2026-07-31

## Document Control

| | |
|---|---|
| **Document No.** | TS-CB-2210 |
| **Title** | Circuit Breaker Troubleshooting Manual |
| **Equipment Class** | HV/MV power circuit breakers — SF6, air-blast, minimum oil, and vacuum interrupter types |
| **Revision** | 01 |
| **Effective Date** | 2026-07-31 |
| **Audience** | Qualified substation/switchgear maintenance technicians and protection engineers |

## 1. Purpose and Scope

This manual provides a structured troubleshooting reference for high- and medium-voltage power circuit breakers, covering the most common fault categories: failure to close, failure to trip, SF6 gas system alarms, operating mechanism faults, and control/protection circuit issues.

This manual supports diagnosis at the level of “what is the probable cause and what should I check next” — it does not replace the OEM instruction manual for a specific breaker model, nameplate ratings, or the site's protection and isolation procedures, all of which take precedence over this document.

## 2. Circuit Breaker Types Covered

Troubleshooting approach differs meaningfully by interrupting medium. Confirm which type you're working on before applying any diagnostic table in this manual.

| Type | Key Characteristics |
|---|---|
| SF6 (gas-insulated) | Sulphur hexafluoride gas provides both insulation and arc interruption. Common in dead tank and gas-insulated switchgear (GIS) designs up to 550 kV. Sensitive to gas pressure/density — low SF6 density is a common alarm/lockout condition. |
| Air-blast | Compressed air interrupts the arc. Requires a reliable, adequately pressurized air supply system; air leaks and compressor issues are common fault sources on older installations. |
| Minimum oil | A small, controlled oil volume interrupts the arc. Largely legacy technology; oil level, dielectric condition, and contact wear are the primary maintenance |

| Type | Key Characteristics |
|---|---|
| | concerns. |
| Vacuum interrupter | Contacts separate inside a sealed vacuum bottle. Common at medium-voltage levels. Loss of vacuum integrity is a critical, non-repairable failure mode requiring interrupter replacement. |

## 3. Safety Precautions

**WARNING:** High- and medium-voltage circuit breakers store significant mechanical energy (charged springs, hydraulic/pneumatic accumulators) even when electrically isolated. Unexpected mechanical release can cause severe injury or death. Never work inside a breaker mechanism compartment without first discharging or blocking stored energy per the OEM manual.

- Do not begin work until a valid switching order/permit has been issued, the breaker is confirmed open, isolated, and grounded per site procedure, and lockout/tagout is applied.
- Treat all control and trip circuits as energized until verified de-energized with a calibrated meter — a breaker can be commanded to close or trip remotely by protection or SCADA systems.
- Discharge or mechanically block closing springs, and vent/block hydraulic or pneumatic accumulators, before any mechanism work, per the OEM procedure.
- Treat SF6 gas compartments with caution: SF6 itself is not toxic, but arc byproducts (decomposition compounds) inside a breaker tank are toxic and corrosive. Use appropriate PPE and ventilation, and follow the site's SF6 handling procedure before opening any gas compartment.
- Use appropriate arc-flash PPE per the site's arc-flash study/label for any work near energized conductors or during switching operations.
- Never bypass a protection or interlock function to “make a test pass” without following the site's management of change (MOC) and authorized bypass process.

## 4. Required Tools and Test Equipment

- Multimeter and/or micro-ohmmeter (contact resistance testing)
- Circuit breaker timing analyzer (open/close timing and contact travel)
- SF6 gas density/pressure gauge and leak detector, where applicable
- Insulation resistance (megohmmeter) tester
- Trip/close coil current signature test set
- Control schematic/wiring diagram and OEM instruction manual for the specific breaker
- Lockout/tagout kit and grounding equipment per site procedure
- Appropriate arc-flash and electrical PPE per the site's arc-flash study

## 5. General Troubleshooting Methodology

1. **Gather the event data.** Record what happened: alarm/annunciation text, SCADA event log, protection relay targets, and whether the breaker was commanded to operate or failed spontaneously.
2. **Confirm breaker status.** Verify actual breaker position (open/closed) using mechanical indicators, not just control room status, since indication faults are themselves a possible root cause.

3. **Isolate and ground.** Follow the site switching order to isolate and ground the breaker before any hands-on diagnostic work requiring compartment access.
4. **Check the obvious first.** Verify control power, SF6/air pressure, and any active lockout or block-close conditions before assuming a component failure.
5. **Consult the relevant table.** Use the troubleshooting tables in Sections 6–9 to narrow the probable cause based on the specific symptom.
6. **Test before replacing.** Confirm a suspected component has actually failed (coil resistance, contact continuity, mechanism travel) before replacing it — misdiagnosis is a common cause of repeat failures.
7. **Document and close out.** Record the fault, root cause, and corrective action in the equipment history, and close the work permit.

## 6. Troubleshooting: Breaker Fails to Close

| Symptom | Probable Cause | Corrective Action |
|---|---|---|
| No response to close command; no indication of mechanism motion | Loss of control power, blown close-circuit fuse, or open close coil | Verify control power at the cabinet; check close-circuit fuses/breakers; measure close coil resistance against OEM spec. |
| Mechanism attempts to close but stops partway (SF6 breakers) | Low SF6 gas density has activated a block-close lockout | Check SF6 density gauge/alarm status; do not force a close operation; restore gas pressure per OEM procedure before re-attempting. |
| Close command given but mechanism does not move at all (spring-operated) | Closing spring not charged, or spring-charge motor failure | Verify spring-charged indication; check spring motor control fuse and motor continuity; manually verify spring charge per OEM procedure if safe to do so. |
| Close command given but anti-pump/interlock prevents closing | An active interlock (e.g., disconnect not fully closed, safety interlock, protection lockout relay) is blocking the close circuit | Trace the control schematic to identify which interlock contact is open; verify the interlocking condition is actually satisfied before resetting. |
| Breaker closes but immediately trips (“close-and-trip”) | An active protection trip condition, or a persistent fault on the protected circuit | Do not re-close without confirming the protection target and investigating the downstream circuit for an actual fault condition. |

## 7. Troubleshooting: Breaker Fails to Trip / Open

**WARNING:** A breaker that fails to trip on a genuine fault condition is a critical safety and reliability issue. Escalate immediately per Section 10 if a trip failure occurs during an actual fault, rather than continuing routine troubleshooting.

| Symptom | Probable Cause | Corrective Action |
|---|---|---|
| No response to trip command from either protection or manual trip | Loss of DC control/trip power, blown trip-circuit fuse, or open trip coil | Verify trip circuit DC supply and fuses; measure trip coil resistance against OEM spec; check trip circuit supervision relay/alarm if fitted. |
| Protection relay issues a | Failed trip coil, seized/binding | Perform a manual/local trip test if safe and |

| Symptom | Probable Cause | Corrective Action |
| --- | --- | --- |
| trip, but breaker does not open | mechanism, or a failed auxiliary trip contact | permitted; check trip coil and auxiliary contact continuity; inspect mechanism for binding per OEM procedure. |
| Breaker trips slowly or with abnormal timing | Mechanism lubrication/binding issues, degraded operating springs, or a weak trip coil | Run a timing test and compare to OEM specification; inspect and service the mechanism per the OEM maintenance schedule if timing is out of tolerance. |
| Local manual trip works but remote/protection trip does not | Wiring or contact fault specific to the remote trip path (relay output contact, wiring, terminal) | Trace the remote trip path on the schematic; check the protection relay's trip output contact and associated wiring/terminals. |

## 8. Troubleshooting: SF6 Gas System

| Symptom | Probable Cause | Corrective Action |
| --- | --- | --- |
| Low SF6 density alarm (Stage 1) | Slow gas leak, or temperature-related density reading near threshold | Log the reading and trend; schedule leak detection survey; top up gas per OEM procedure if confirmed low, not just temperature-related. |
| Low SF6 density lockout (Stage 2 / block-close) | Gas density has dropped below the safe operating threshold for interruption | Do not attempt to operate the breaker; treat as an urgent condition; isolate, investigate the leak source, and restore gas pressure per OEM procedure before any operation. |
| Audible gas leak or visible frost/condensation at a fitting | Failed seal, loose fitting, or valve leak | Follow site SF6 handling/PPE procedure; use a calibrated leak detector to confirm and locate the leak before any repair attempt. |
| SF6 moisture/dew point alarm | Moisture ingress, often from a compromised seal or desiccant saturation | Follow OEM procedure for gas analysis; do not disregard — moisture increases risk of internal flashover and byproduct formation. |

## 9. Troubleshooting: Operating Mechanism and Control Circuit

| Symptom | Probable Cause | Corrective Action |
| --- | --- | --- |
| Spring-charge motor runs continuously or fails to stop | Failed limit switch or charge-indication contact | De-energize the motor circuit per LOTO; inspect and test the charging limit switch per OEM procedure. |
| Hydraulic mechanism low-pressure alarm | Internal leak, pump/motor fault, or temperature-related pressure drop | Check pump run frequency and pressure trend; investigate for external/internal leaks per OEM procedure before topping up fluid. |
| Air-blast mechanism low | Compressor fault, air leak in the | Check compressor operation and supply |

| Symptom | Probable Cause | Corrective Action |
|---|---|---|
| air pressure | supply system, or a stuck drain/relief valve | line for leaks; verify drain valves are not stuck open; restore pressure before operating. |
| “Control circuit faulty” or supervision alarm active | An open circuit somewhere in the trip or close supervision loop (commonly a wiring or terminal fault) | Trace the supervision circuit on the schematic; check terminal connections and continuity segment by segment. |
| Breaker position indication disagrees with actual mechanical position | Failed or misaligned auxiliary switch (52a/52b contacts) | Visually verify actual mechanical position first; inspect and adjust/replace the auxiliary switch per OEM procedure. |

## 10. Escalation Criteria

Stop routine troubleshooting and escalate to engineering, the OEM, or a specialized service provider when any of the following apply:

- A breaker fails to trip during an actual fault condition (safety-critical failure).
- SF6 gas has reached a lockout-level low density and the leak source cannot be quickly identified and isolated.
- Internal arcing, unusual odor, discoloration, or visible damage is found inside a gas or oil compartment.
- Timing test results are significantly out of tolerance and the cause is not resolved by routine adjustment/lubrication.
- A vacuum interrupter is suspected of having lost vacuum integrity (this is a non-repairable, replace-only condition).
- Root cause cannot be isolated after following the applicable tables in this manual, and the breaker remains out of service.

## 11. Revision History

| Rev. | Date | Author | Description of Change |
|---|---|---|---|
| 00 | 2026-03-02 | M. Langley, Tech. Writer | Initial release covering fail-to-close and fail-to-trip conditions. |
| 01 | 2026-07-31 | M. Langley, Tech. Writer | Added SF6 gas system, operating mechanism, and control circuit troubleshooting sections; added escalation criteria. |